

8. Retrieval-Augmented Generation (RAG) Experiments and Evaluation

8.1 Motivation for RAG Integration

While fine-tuned Small Language Models (SLMs) demonstrated strong performance on multilingual sentiment and language classification tasks, they remain constrained by AGen due to limited parameter capacity and lack of up-to-date or domain-specific knowledge. To address these limitations, **Retrieval-Augmented Generation (RAG)** was introduced as an auxiliary framework to enhance response quality by grounding model outputs in externally retrieved contextual documents.

The primary goal of integrating RAG was to:

- Improve **answer relevance** for user queries
- Reduce hallucinations by enforcing **context grounding**
- Evaluate how well small models utilize retrieved knowledge

8.2 RAG System Architecture

The implemented RAG pipeline consists of the following components:

1. **Query Encoder**
User queries are encoded into dense vector representations using a lightweight embedding model optimized for retrieval efficiency.
2. **Document Store**
A vector database stores chunked and embedded documents derived from curated textual sources relevant to the task domain.
3. **Retriever**
Top-k relevant document chunks are retrieved using cosine similarity between query and document embeddings.
4. **Generator (SLM)**
The retrieved context is appended to the user query and passed to the fine-tuned SLM, which generates a grounded response.

This modular design enables scalability while remaining computationally feasible for resource-constrained environments.

8.3 Evaluation Methodology

To evaluate the effectiveness of the RAG system, three standard RAG-specific metrics were employed:

Metric	Description
Answer Relevancy	Measures how well the generated answer addresses the user query
Context Precision	Evaluates whether retrieved documents are relevant to the query

Metric	Description
Faithfulness	Measures how strictly the generated answer adheres to the retrieved context

All metrics were computed across a representative evaluation set using automated RAG evaluation pipelines.

8.4 Quantitative Results

The RAG evaluation yielded the following results:

Metric	Score
Answer Relevancy	0.73
Context Precision	1.00
Faithfulness	0.38

8.5 Analysis of Results

8.5.1 Answer Relevancy (0.73)

The answer relevancy score indicates that the RAG-enhanced model generally produces responses that align well with user intent. A score of 0.73 reflects **strong semantic understanding**, particularly given the compact size of the underlying language model. This demonstrates that retrieval-augmented context significantly improves response usefulness compared to standalone generation.

8.5.2 Context Precision (1.00)

The perfect context precision score highlights the **high effectiveness of the retrieval mechanism**. Retrieved documents were consistently relevant to the input queries, confirming:

- High-quality document chunking
- Effective embedding representations
- Accurate similarity-based retrieval

This result validates the robustness of the retrieval pipeline independent of the generation component.

8.5.3 Faithfulness (0.38)

The relatively low faithfulness score indicates that the model **does not always strictly ground its responses in the retrieved context**, occasionally introducing unsupported or inferred information. This behavior is common in small language models and can be attributed to:

- Limited instruction-following capability
- Over-generalization during generation

- Insufficient grounding constraints in the prompt template

Despite accurate retrieval, the generator sometimes relies on internal priors rather than the provided evidence.

8.6 Key Observations

- The **retrieval component outperforms the generation component**, as seen by high context precision but lower faithfulness.
- RAG significantly improves **answer relevance**, even when strict grounding is imperfect.
- Small Language Models benefit from RAG, but require **explicit grounding enforcement** to minimize hallucinations.
- Faithfulness remains the primary bottleneck in SLM-based RAG systems.

8.7 Limitations

- The generator occasionally paraphrases or extrapolates beyond retrieved context.
- Limited model capacity restricts nuanced reasoning over long retrieved passages.
- Prompt templates were not explicitly optimized for faithfulness enforcement.

8.8 Future Improvements

To improve RAG performance in future iterations, the following enhancements are recommended:

1. **Faithfulness-aware Prompt Engineering**
Explicitly instruct the model to answer *only* using retrieved context.
2. **Answer-Context Attribution Training**
Fine-tune the model on examples that penalize unsupported statements.
3. **Smaller Context Windows with Higher Relevance**
Reduce noise by aggressively filtering retrieved passages.
4. **Post-Generation Faithfulness Re-ranking**
Introduce a secondary verifier model to reject ungrounded responses.

8.9 Conclusion

The RAG experiments demonstrate that **Retrieval-Augmented Generation is a highly effective enhancement for Small Language Models**, particularly in improving answer relevance and reducing dependency on parametric knowledge. While retrieval accuracy is excellent, ensuring faithful generation remains an open challenge. Overall, this work validates RAG as a practical and scalable approach for deploying efficient, knowledge-grounded AI systems under resource constraints.